

TURN-BY-TURN TUNE ANALYSIS USING ADAPTIVE BPM ENSEMBLES IN THE FERMILAB Mu2e DELIVERY RING*

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Abstract

The Mu2e experiment at Fermilab requires stable resonant slow extraction from the Delivery Ring, making reliable tune monitoring an important operational diagnostic. This work investigates BPM-based tune-candidate extraction using synchronized turn-by-turn position data distributed across multiple digitizers. Each spill contains approximately 50,000 turns from many BPMs in both transverse planes, enabling spectral analysis of tune-like structure.

The analysis captures coherent spill snapshots, verifies synchronization using stream timestamps, and computes tune candidates in configurable horizontal and vertical tune bands. Rather than relying on a single BPM or fixed BPM list, it evaluates BPM quality on a spill-by-spill basis and selects small adaptive BPM ensembles. A multi-spill study shows that tune observability is distributed and dynamic rather than concentrated in one globally optimal BPM. Adaptive ensembles improve tune-candidate quality compared with single-BPM selections, with the clearest results in the vertical plane. The horizontal plane shows useful ranking structure but weaker visibility under present thresholds.

Direct evaluation of fixed global BPM sets shows that static selections do not reproduce dynamic per-spill performance. These results motivate an adaptive BPM-ensemble approach for Delivery Ring tune analysis using selected BPM subsets, confidence metrics, and quality flags rather than a single preferred BPM or fixed BPM list.

INTRODUCTION

The Mu2e slow-extraction programme uses third-integer resonant extraction from the Fermilab Delivery Ring. Spill regulation depends on a tune-ramp quadrupole family and radio-frequency knock-out excitation, making transverse tune observability relevant to both commissioning and operations [1–3]. The installed Muon Campus BPM system provides synchronized turn-by-turn (TBT) position waveforms throughout the ring [4]. A single BPM, however, need not remain the best observer as orbit, oscillation amplitude, extraction conditions, and instrument quality change.

Combining TBT data from multiple BPMs can improve tune estimation when individual signals have unequal signal-to-noise ratio or decohere rapidly [5, 6]. The central question here is therefore not which BPM is globally best,

but how many BPMs should be combined for each spill and whether that adaptive choice is reproducible outside the data used to make it.

This paper reports a BPM-only internal validation. The reported values are fractional tune candidates in configured search bands. No Schottky, tune-meter, or controlled tune-knob reference was available for these captures, so the analysis does not claim absolute tune accuracy.

DATA AND SYNCHRONIZATION

The primary data set contains 2000 captured spills in two acquisition collections. Its nominal topology supplies 60 horizontal and 60 vertical position channels, with two same-plane sources per digitizer; 12 flagged partial captures contain 16 recorded source absences. Absent channels remain missing rather than being fabricated or zero-filled. Each recorded waveform contains approximately 50,000 turns. The analysis consumes raw host-published position arrays: “raw” means DSP-derived position floats before host-side threshold substitution and scaling, not electrode ADC samples. Stream timestamps are used to assemble a coherent spill snapshot. Adjacent target buckets are merged within 1 ms, while captured streams are checked against the selected target using the separately configured 25 ms same-spill tolerance. Incomplete or inconsistent states carry quality flags rather than being silently accepted.

Publication auditing exposed two identity defects in an earlier analysis. First, downstream tables stored a shared digitizer label rather than the exact channel key, even though two same-plane channels could share that label. Second, ring order was parsed from the first number in the digitizer address, assigning every channel the same order and disabling the intended ring-span term. The corrected pipeline serializes plane-local indices, exact BPM tokens, full source keys, and digitizer identities. Every downstream reconstruction resolves the subset bit mask before display labels.

The waveform path subtracts the window mean, applies a Hann taper, and computes a zero-padded real fast Fourier transform. Frequencies are mapped to the upper fractional tune branch by $q = 1 - f$. The discovery bands are $0.60 \leq q_x \leq 0.70$ and $0.67 \leq q_y \leq 0.75$; expected values near 0.65 and 0.72 enter only as soft priors.

ADAPTIVE ENSEMBLE METHOD

For the primary search, overlapping 4,096-turn windows with 256-turn stride characterize early-spill spectral structure. Per-BPM peak candidates are clustered within each spill and plane to form an internal consensus; no distribu-

* This manuscript has been authored by Fermi Forward Discovery Group, LLC under Contract No. 89243024CSC000002 with the U.S. Department of Energy, Office of Science, Office of High Energy Physics.

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tion across other spills is used as a label. Candidate subset spectra are averaged in power. The score combines held-out channel support, robust peak prominence and entropy, agreement with the within-spill consensus, window stability, visibility, digitizer and ring diversity, and an ambiguity penalty. Turn-dependent diagnostics are retained because extraction records are nonstationary [7].

Best-1 and Best-3 are searched exhaustively over the 60 channels in a plane. Best-5 uses exhaustive enumeration within a recorded 20-channel screening pool. Because the score is a model-selection statistic rather than a measurement uncertainty, all publication claims are checked in the independent protocol below.

Leakage-controlled Best- N selection

The ensemble-size study sweeps contiguous N values instead of comparing only 1, 3, and 5. Members are selected from a prefix of eight fit windows. Every later window overlapping that prefix is purged; evaluation begins at the first non-overlapping window. Candidate tunes are derived only from fit windows.

The full N curve uses 4000 spill-plane cases; digitizer-disjoint validation uses an evenly stratified 1000-case subset, each evaluated across 5 folds rather than being presented as the full curve population.

A bounded beam search is used above $N = 1$ and is repeated at multiple beam widths. Convergence is judged from membership, score, and tune differences, not membership alone. For channel-disjoint validation, complete digitizers are assigned to folds so sibling channels never appear on opposite sides. Each fold selects from the training digitizers and is evaluated on later windows; the reference is a median-power spectrum from held-out digitizers.

Two agreement measures are retained. The conditioned measure searches near the training tune and answers whether later spectra support that candidate. The blind measure searches the full configured band independently for selected and held-out channels. The smallest non-inferior N is chosen only after blind agreement and channel-disjoint tune difference are screened, followed by both selected-channel and held-out-channel power support and prominence. Confidence intervals collapse folds within a spill and use a moving-block bootstrap within each capture collection to retain short-range serial dependence. As a final hyperparameter check, N is chosen from one acquisition collection and the same global N is evaluated in the other; only the per-spill members remain adaptive.

RESULTS

Corrected adaptive search

Across 4,000 corrected spill-plane rows per subset size, the median adaptive-search score rises from 0.343 to 0.392 and 0.403 for horizontal Best-1, Best-3, and Best-5, and from 0.420 to 0.484 and 0.524 vertically. Paired Best-1-to-Best-3 gains are 0.0500 horizontally and 0.0646 vertically;

Best-3-to-Best-5 gains are 0.0122 and 0.0310. These are in-sample model-selection scores, not tune-accuracy measurements. Leakage-controlled later-window results and exact-paired persistence metrics are reported in Table 1.

Across seven verified reduced-sample beam, fit-window, and fold-seed checks, eligible H knees span Best-2–Best-13 in 5/7 runs, with 2 unresolved; V spans Best-10–Best-28 in 6/7, with 1 unresolved. Unresolved cases are retained as sensitivity evidence rather than coerced into a recommendation; the full-data selections drive the persistence figures.

Full-spill ridge-density comparison

To visualize persistence through the approximately 50,000-turn record, the selected early-window membership is applied to the raw captured spill. Each selected trace is RMS-normalized, 4,096-turn Hann spectra are computed every 256 turns, and selected-BPM powers are averaged. One locally tracked ridge pick per spill and window is binned across spills. This plot is a density of selected ridge locations, not a spectral-power heat map. Of 360000 structural rows per plane, H retains 359018 finite in-band picks (14 blank and 968 bounded edge exclusions), while V retains 289210 (69684 blank and 1106 edge exclusions); none is imputed.

The legacy comparator uses identical tune bands and binning but came from a normalized single-channel selector whose winner was effectively determined by floating-point residuals after RMS normalization. It is therefore labeled “legacy normalized-single”, not “best BPM”. Paired panels use exact common spill/window ridge points and one color scale. Difference maps subtract column-normalized ridge-pick densities; they quantify redistribution and concentration of ridge picks, not physical noise removal. A separate corrected-Best-1 comparison isolates ensemble size from repair of that historical selector. Moving-block intervals across turn centers test whether paired concentration changes persist despite the heavily overlapping windows; they are not uncertainty estimates over the spill population.

Fixed sets, intensity, and plane asymmetry

Frozen global top-1, top-3, and top-5 sets are trained on one acquisition collection and applied to the other. Dynamic memberships, frozen memberships, and all-BPM controls are rescored from the same cached spectra with one evolution metric. This comparison is descriptive because the original dynamic memberships reuse selection windows; the corrected result is reported alongside the leakage-controlled Best- N inference rather than used as an independent validation test.

Dynamic Best-5 exceeds frozen Best-5 vertically under this descriptive score, but all-BPM aggregation remains strong. A second, leakage-controlled comparison therefore aggregates every training-side channel by mean and median under the same purged later windows and held-out-digitizer folds as Best- N . Of eight method–metric comparisons per plane, selected Best- N is favored in 2 H and 3 V comparisons, all-training aggregation is favored in 3 H and 3 V

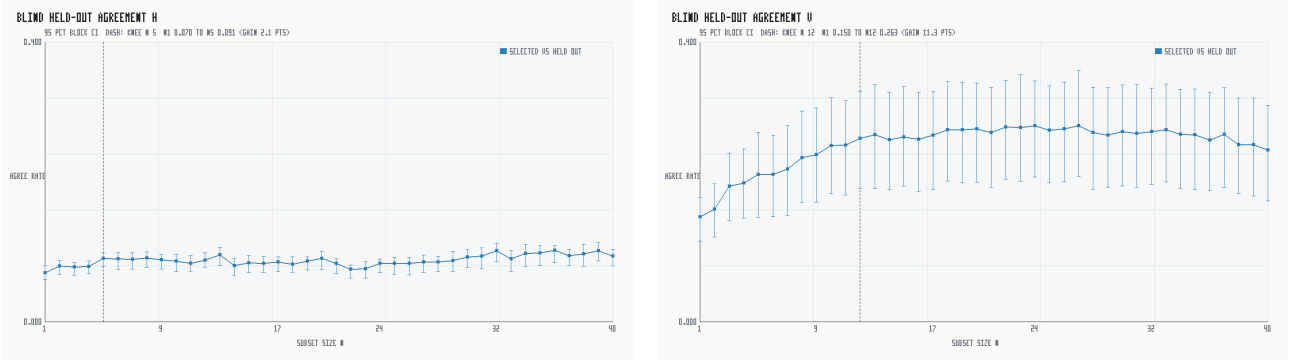


Figure 1: Leakage-controlled Best- N validation for the horizontal (left) and vertical (right) planes. Blind full-band selected-versus-held-out digitizer agreement is shown with moving-block intervals; conditioned support near the fitted candidate is reported separately.

Table 1: Leakage-controlled Best- N intervals use collection-preserving spill blocks. Ridge intervals use overlapping-turn blocks on exact-paired picks and describe concentration, not absolute tune accuracy or measured physical noise.

Plane	Best- N	Blind agreement	Blind $ \Delta q $	ΔIQR vs B1	ΔIQR vs legacy
H	5	0.091 [0.079, 0.098]	0.0161 [0.0155, 0.0172]	-0.0024 [-0.0031, -0.0019]	-0.0038 [-0.0057, -0.0031]
V	12	0.263 [0.191, 0.33]	0.0107 [0.0097, 0.012]	0.001 [3.87e-04, 0.0017]	3.99e-05 [-9.92e-04, 5.98e-04]

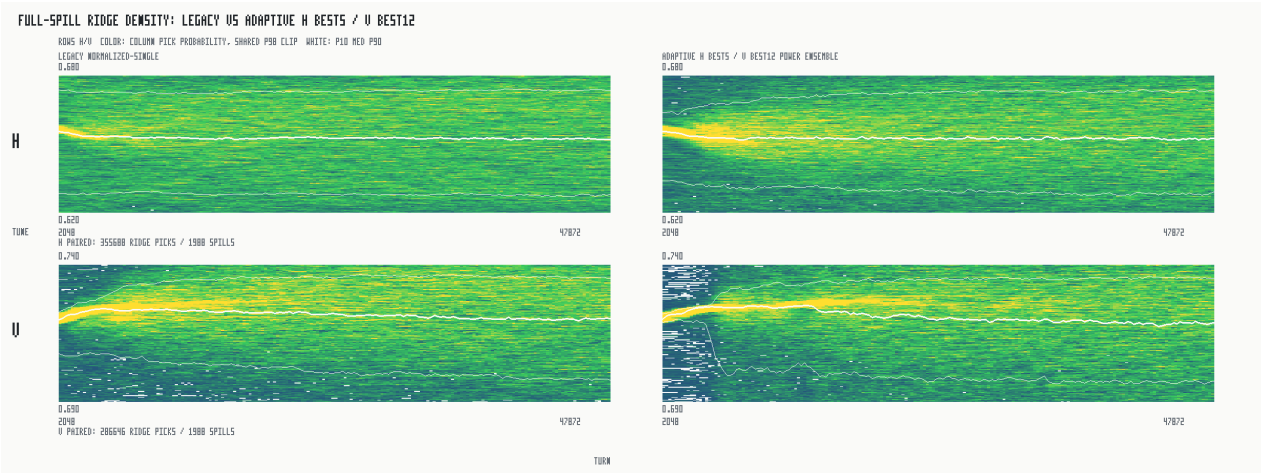


Figure 2: Full-spill ridge-pick density for the legacy normalized-single method and the selected adaptive ensemble on exact common spill/window points. Color is column-normalized cross-spill pick probability with one shared P98-clipped scale; white curves show cross-spill P10, median, and P90 tracks. Horizontal and vertical results are interpreted separately.

comparisons, and 3 H plus 2 V comparisons remain unresolved. This is an internal reproducibility control, not an external tune calibration.

A separate 200-spill capture supplies 23,999 exact raw position/intensity channel pairs; one expected pair is absent. For the pairs present, the first 50,000 turns pass the structural checks, while advertised longer arrays become unreliable near turn 64,000 and are excluded. Square-root, linear, and gated intensity weights are compared with unweighted spectral aggregation on purged later windows. A window with no usable selected intensity falls back explicitly to unweighted aggregation, and the $N = 1$ case is an exact zero-effect control. Across 240 paired method-effect tests, a 20-spill moving-block bootstrap and block sign-flip test leave 0 false-discovery-rate-significant directional effects within tune tolerance, 0 effects above the declared practical thresh-

olds, and 0 retained weighting effects. Intensity is therefore retained as an envelope and integrity diagnostic rather than a tune weight.

The vertical plane gives the clearest sustained, channel-disjoint structure. Horizontal rankings improve with adaptive selection, but visibility decays or becomes diffuse earlier and more variably. No fixed extraction-onset turn is imposed: the machine onset can vary, and a previously suggested 10,000–20,000-turn interval is treated only as review context. Horizontal loss diagnostics therefore report turn-dependent concentration and data-derived sustained-loss points without assigning cause.

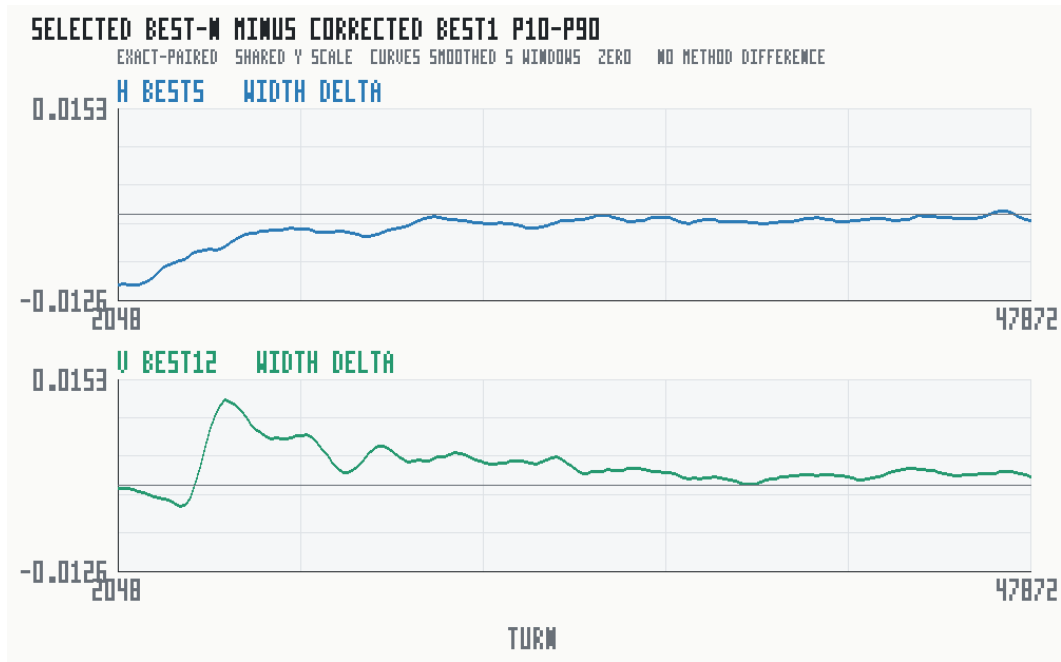


Figure 3: Exact-paired selected-Best- N minus corrected-Best-1 P10–P90 ridge-pick width; negative values indicate a narrower selected-ensemble distribution. Panels share one y scale, and curves are five-window visual smooths of unsmoothed exports. This isolates ensemble size, not physical noise removal, absolute tune accuracy, or extraction timing.

LIMITATIONS AND OPERATIONAL USE

The study has four principal limits. First, internal BPM agreement is not an independent tune calibration. Second, unlabeled machine states limit block-aware uncertainty; controlled scans remain necessary. Third, fixed early-window membership makes the 50,000-turn visualization a persistence test, not full-spill reselection. Fourth, ridge concentration can improve if all methods follow the same biased candidate. A matched Schottky or tune-meter reference and controlled tune-knob study are therefore required.

Within those limits, adaptive ensembles are preferable to a single BPM or static list when channel identity and quality diagnostics matter. All-training aggregation remains an explicit same-protocol control. An operational monitor should expose exact members, ensemble size, tune candidate, prominence and ambiguity, later-window support, and a no-reliable-tune state, with plane-independent quality decisions.

CONCLUSION

Adaptive BPM ensembles recover repeatable tune-like structure. Leakage-controlled, digitizer-disjoint validation selects a useful size knee relative to Best-1; vertical evidence is strongest and horizontal tracking remains limited. Frozen sets and intensity weighting do not replace adaptation, while same-protocol all-training outcomes remain explicit. This BPM-only diagnostic still requires external tune calibration.

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